

PROBLEM SET 5—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions: These are my suggested solutions. It is your responsibility to check this document against your own solutions.

1. (a) Let D be a single-fixed repayment scheme. As seen in the lecture notes, type- i borrowers apply for funds if and only if $D \leq R_i$. If $D > R_s$ only the risky borrowers will consider applying. The maximum then that the bank can charge is $D = R_r$. In this case, the bank is only contracting with risky borrowers and obtains $(1 - \beta)(p_r R_r - 1)$. In contrast with the case analyzed in class, assumption A1' implies that this term is negative. Thus, contracting with risky borrowers only is not optimal. This implies that $D \leq R_s$, i.e. the bank should contract with both types, assuming of course that this is profitable. If this is the case, the bank should charge the highest possible repayment that the safe borrowers can afford, that is $D = R_s$. In this case, the profits obtained by the bank are given by

$$\alpha(\beta(p_s R_s - 1) + (1 - \beta)(p_r R_s - 1)).$$

In order for the bank to be willing to contract with both borrowers, the expression above must be positive. Thus, profits are positive if and only if $R_s(\beta p_s + (1 - \beta)p_r) \geq 1$ and in this case the bank finds optimal to contract with both types. On the other hand, if $R_s(\beta p_s + (1 - \beta)p_r) < 1$ the bank abstains from lending to any of the borrowers. Note that in both situations, there is credit rationing.

- (b) We need to characterize the second-best contract for the bank. A contract is defined by (x_i, D_i) where x_i is the probability of obtaining financing and D_i is the amount to be repaid. The bank's problem is:

$$\max_{(x_i, D_i)_{i=s,r}} \{\beta x_s(p_s D_s - 1) + (1 - \beta)x_r(p_r D_r - 1)\} \quad (1)$$

$$\text{subject to } 0 \leq x_i \leq 1, \quad \forall i = s, r \quad (2)$$

$$\beta x_s + (1 - \beta)x_r \leq \alpha \quad (3)$$

$$x_i p_i (R_i - D_i) \geq 0 \quad \forall i = s, r \quad (4)$$

$$x_i p_i (R_i - D_i) \geq x_j p_i (R_i - D_j) \quad \forall i, j = s, r \quad (5)$$

Because of assumption A2, the risky type still has the highest willingness to repay, so it is still considered the “high type”. Therefore, the previous problem is equivalent to the reduced problem:

$$\max_{(x_i, D_i)_{i=s,r}} \{\beta x_s(p_s D_s - 1) + (1 - \beta)x_r(p_r D_r - 1)\} \quad (6)$$

$$\text{subject to } x_s p_s (R_s - D_s) \geq 0 \quad (7)$$

$$x_r p_r (R_r - D_r) \geq x_s p_r (R_r - D_s) \quad (8)$$

$$x_r \geq x_s \quad (9)$$

$$0 \leq x_i \leq 1 \quad i = s, r \quad (10)$$

$$\beta x_s + (1 - \beta)x_r \leq \alpha \quad (11)$$

Furthermore, we know that the participation constraint for the safe type and the incentive constraint for the risky type will be binding. Solving for D_s and D_r using these binding constraints and plugging the resultant expressions into the objective yields the following problem:

$$\max_{x_s, x_r} x_s (\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s)) + x_r((1 - \beta)(p_r R_r - 1)) \quad (12)$$

$$\text{subject to } 0 \leq x_s \leq x_r \leq 1 \quad (13)$$

$$\beta x_s + (1 - \beta)x_r \leq \alpha \quad (14)$$

Notice that in the objective (12), the term attached to x_r is strictly negative. Thus the bank would like to shutdown any interaction with the risky types. However, if $x_r = 0$ then it would force x_s to be zero also because of the monotonicity constraint. Since it might still be profitable to contract with the safe type, then $x_s = 0$ might not always be optimal. This implies that the monotonicity constraint will be binding. To see this, suppose towards a contradiction that $\hat{x}_s$ and $\hat{x}_r$ are optimal solutions such that $\hat{x}_r = \hat{x}_s + \epsilon$ where $\epsilon > 0$. The expected profit then is

$$\hat{x}_s [R_s(\beta p_s + (1 - \beta)p_r) - 1] + \epsilon(1 - \beta)(p_r R_r - 1).$$

However, by taking instead $\tilde{x}_r = \hat{x}_s$ we have that the profit from the contract $(\hat{x}_s, \tilde{x}_r)$ is such that

$$\hat{x}_s [R_s(\beta p_s + (1 - \beta)p_r) - 1] + \epsilon(1 - \beta)(p_r R_r - 1) < \hat{x}_s [R_s(\beta p_s + (1 - \beta)p_r) - 1]$$

which contradicts the optimality of $\hat{x}_r$. Thus, the monotonicity constraint must be binding.

Let $x = x_r = x_s$. Then the problem reduces to

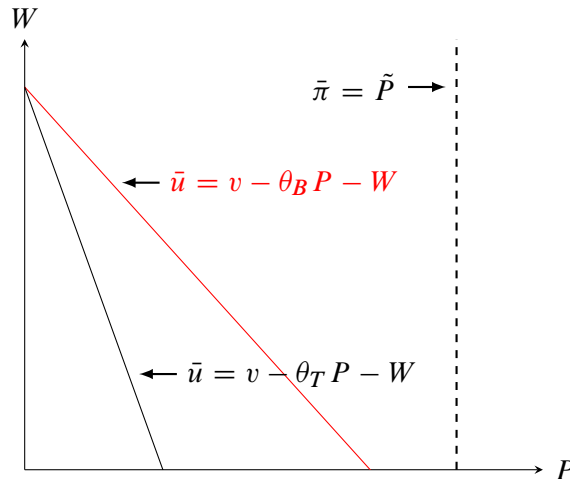
$$\max_{x \in [0, \alpha]} x [R_s(\beta p_s + (1 - \beta)p_r) - 1] \quad (15)$$

From the previous problem it is easy to see that we have the following optimal solution

$$x^* = \begin{cases} \alpha & \text{if } R_s(\beta p_s + (1 - \beta)p_r) \geq 1, \\ 0 & \text{if } R_s(\beta p_s + (1 - \beta)p_r) < 1. \end{cases}$$

With this solution, it is easy to find the optimal repayment scheme from the binding incentive and participation constraints: $D_r = D_s = D^* = R_s$. Thus, the optimal contract is a *pooling contract* and it is the same as the one found in the previous item. Furthermore, the solution implies that if the bank decides to actually fund any of the consumers, it is cross-subsidizing the risky types with the profits obtained from the safe ones. If this is the case, there is still credit rationing because the probability of funding any type of consumer is α . When the bank decides not to contract with any type, there is a total financial collapse, since no consumer gets credit.

2. (a) The following diagram plots the indifference curves of the two types of travelers and the iso-profit line of the airline in (P, W) -space.



The airline only pays cost c of a passenger if it chooses to provide service, i.e. if it chooses to charge at least one strictly positive price. If by charging at least one strictly positive price the airline finds that it cannot make a profit, then it decides not to operate. Thus, the problem of the airline is

$$\begin{aligned}
& \max_{\{(P_T, W_T), (P_B, W_B)\}} \max\{\lambda P_T + (1 - \lambda)P_B - c, 0\} & (16) \\
& \text{subject to } \theta_B P_B + W_B \leq \theta_B P_T + W_T & (\text{IC B}) \\
& \theta_T P_T W_T \leq \theta_T P_B + W_B & (\text{IC T}) \\
& v \geq \theta_B P_B + W_B & (\text{PC B}) \\
& v \geq \theta_T P_T + W_T & (\text{PC T}) \\
& P_T \geq 0, P_B \geq 0, W_T \geq 0, W_B \geq 0 & (\text{Non-Negativity})
\end{aligned}$$

The airline's problem is written in that way because if it decides not to operate, then it doesn't need to pay a cost. The airline doesn't operate when it charges both prices equal to zero. In order to simplify this problem, we will assume for the moment that the airline decides to operate always, i.e. that profit is always positive. Thus we will consider as an objective

$$\lambda P_T + (1 - \lambda)P_B - c,$$

and we will worry later about in what circumstances the firm will make negative profits.

- (b) In the maximization problem above we have that (PC T) and (IC B) together with $\theta_B < \theta_T$ (which in this problem is the single-crossing property) imply that constraint (PC B) is redundant, so it is never binding. constraint (PC T) must therefore be binding for if it would not, then we can reduce P_T and P_B by a small positive quantity $\epsilon > 0$ and all the remaining constraints will still be satisfied and this would improve the airline's profit. The fact that (PC T) is binding implies that tourists are indifferent between buying and not buying a ticket.
- (c) Suppose that $\{(P_T, W_T), (P_B, W_B)\}$ is an optimal, incentive compatible contract. We need to show that $W_B = 0$. Towards a contradiction, assume that $W_B > 0$. Choose $\epsilon > 0$ to reduce W_B by ϵ such that it still remains positive and increase P_B by ϵ/θ_B so that the utility of type B does not change. Now the firm earns higher profits from the business type. The actual change in profit is $(1 - \lambda)\epsilon/\theta_B > 0$. Now we need to check that this new contract we constructed is not attractive for the tourist type. Indeed this is the case:

$$\theta_T P_T + W_T \leq \theta_T P_B + W_B = \theta_T \left(P_B + \frac{\epsilon}{\theta_B} \right) + (W_B - \epsilon) < \theta_T \left(P_B + \frac{\epsilon}{\theta_B} \right) + (W_B - \epsilon),$$

where the first inequality comes from $\{(P_T, W_T), (P_B, W_B)\}$ being an incentive compatible plan, the second equality comes from altering W_B by ϵ but then increasing P_B by ϵ/θ_B so as to not alter the utility of the tourist type, and the third inequality comes from the fact that $\theta_B < \theta_T$. The previous calculations shows that the altered contract is still incentive compatible but it achieves a higher profit for the airline, contradicting the optimality of $\{(P_T, W_T), (P_B, W_B)\}$ which we assumed originally. Thus we must have that $W_B = 0$ at an optimal incentive compatible plan.

Now, if at an optimal contract, business travelers were not indifferent between (P_T, W_T) and (P_B, W_B) , we could slightly raise P_B and all the constraints would remain satisfied, since recall that (PC B) is redundant, and the airline would increase its profit coming from business types. Therefore, in an optimal contract, business types are indifferent between (P_T, W_T) and (P_B, W_B) : this is another way of expressing that (IC B) is *binding*, $\theta_B P_B + W_B = \theta_B P_T + W_T$.

- (d) From part (b) and (c), consider the binding constraints:

$$v = \theta_T P_T + W_T \tag{17}$$

$$\theta_B P_B + W_B = \theta_B P_T + W_T \tag{18}$$

These two constraints together with monotonicity ($P_B \geq P_T$) and single crossing, imply that

$$W_T - W_B = \theta_B P_B - \theta_B P_T = \theta_B (P_B - P_T) \leq \theta_T (P_B - P_T)$$

which is the constraint (PC T).

Thus, once again, we can solve the airline's problem by considering just the binding constraints, the monotonicity constraint, and the non-negativity constraints. From the binding constraints, we will solve for the prices in terms of the remaining waiting time:

$$P_T = \frac{v - W_T}{\theta_T} \quad (19)$$

$$P_B = \frac{v - W_T}{\theta_T} + \frac{W_T}{\theta_B} \quad (20)$$

Note that since $P_T \geq 0$ and $P_B \geq 0$ imply from the previous equations that $W_T \leq v$. Plugging in the previous expressions into the objective of the airline yields the following problem:

$$\max_{W_T} \left\{ \lambda \left(\frac{v - W_T}{\theta_T} \right) + (1 - \lambda) \left(\frac{v - W_T}{\theta_T} + \frac{W_T}{\theta_B} \right) - c \right\} \quad (21)$$

$$\text{subject to } 0 \leq W_T \leq v \quad (22)$$

Note that monotonicity of the prices is automatically satisfied from equations (19) and (20).

We can simplify the objective (21) to get:

$$\frac{v - W_T}{\theta_T} + (1 - \lambda) \frac{W_T}{\theta_B} = \left(\frac{v}{\theta_T} \right) - c = \left(\frac{v}{\theta_T} - c \right) + W_T \left(\frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right)$$

Remember that we will eventually need to check if it is worth for the firm to operate, but for the moment, the problem of a firm that decides to operate can be reduced to:

$$\max_{W_T \in [0, v]} \left\{ \left(\frac{v}{\theta_T} - c \right) + W_T \left(\frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right) \right\} \quad (23)$$

If we analyze problem (23) from a purely mathematical perspective, we can notice that the objective function is an *affine function*, i.e. a line with an intercept, and that the maximization is carried out over an interval. Thus, there are only two possible solutions, which depend on the *sign* of the slope $\left(\frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right)$. The solution is given by:

$$W_T = \begin{cases} v & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ 0 & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

Before solving for the remaining variables, we need to check under what circumstances the previous is indeed a *true solution* by checking if the firm is making profits. Thus, we will compute first the profits under the two cases above.

$$\text{Profits when } W_T = 0 : \quad \frac{v}{\theta_T} - c$$

$$\text{Profits when } W_T = v : \quad \frac{(1 - \lambda)v}{\theta_B} - c$$

The firm decides to operate whenever the parameters are such that:

$$\min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} \geq c$$

Whenever the above condition is satisfied, then the remaining variables are given by

$$P_T^* = \begin{cases} 0 & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ \frac{v}{\theta_T} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

$$P_B^* = \begin{cases} \frac{v}{\theta_B} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ \frac{v}{\theta_T} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

If the firm decides not to operate, then all the simplifications above (items b–d) do not apply and the optimal contract is $\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \{(0, 0), (0, 0)\}$

Summarizing all these results, the optimal contract is:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \begin{cases} \{(0, v), \left(\frac{v}{\theta_B}, 0\right)\} & \text{if } \min \left\{ \frac{\theta_T - \theta_B}{\theta_T}, \frac{v - c\theta_B}{v} \right\} \geq \lambda \\ \left\{ \left(\frac{v}{\theta_T}, 0\right), \left(\frac{v}{\theta_T}, 0\right) \right\} & \text{if } \lambda > \frac{\theta_T - \theta_B}{\theta_T} \text{ and } \frac{v}{\theta_T} \geq c \\ \{(0, 0), (0, 0)\} & \text{if } \min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} < c \end{cases} \quad (24)$$

(e) From the optimal solution in (24) we can see that whenever

$$\min \left\{ \frac{\theta_T - \theta_B}{\theta_T}, \frac{v - c\theta_B}{v} \right\} \geq \lambda$$

, the airline is only contracting with the business type:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \left\{ (0, v), \left(\frac{v}{\theta_B}, 0\right) \right\}$$

since the price it charges for tourists is zero. If this is the case, we are in a *shutdown contract*.

On the other hand, whenever

$$\lambda > \frac{\theta_T - \theta_B}{\theta_T} \text{ and } \frac{v}{\theta_T} \geq c$$

the airline optimal plan is a *pooling contract*:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \left\{ \left(\frac{v}{\theta_T}, 0\right), \left(\frac{v}{\theta_T}, 0\right) \right\}.$$

In the case that the firm can never make positive profits, i.e. when

$$\min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} < c,$$

then the optimal contract is also a pooling contract in which no traveler gets serviced.

3. (a) Let $(q(c), P(c))$ be a contract. At the first-best, the principal solves, for any $c \in [\underline{c}, \bar{c}]$

$$\begin{aligned} & \max_{q(c), P(c)} B(q(c)) - P(c) \\ & \text{subject to } P(c) - cq(c) \geq 0. \end{aligned}$$

At an optimal solution, the participation constraint must be binding, i.e., $P(c) = cq(c)$. Substituting the binding constraint into the objective yields, for any $c \in [\underline{c}, \bar{c}]$, the following problem

$$\max_{q(c)} B(q(c)) - cq(c)$$

The first order condition for the above problem is

$$B'(q(c)) = c \quad \forall c \in [\underline{c}, \bar{c}]$$

which characterizes the first-best solution for the principal.

- (b) I will use here a general distribution function $f(c)$ and will specialize to the case of the uniform later. The optimal second-best contract is given by the solution to

$$\begin{aligned} \max_{q(c), P(c)} \quad & \int_{\underline{c}}^{\bar{c}} (B(q(c)) - P(c)) f(c) dc \\ & P(c) - cq(c) \geq 0, \forall c \\ & P(c) - cq(c) \geq P(c') - cq(c'), \forall c, c' \end{aligned}$$

Define $U(c) = P(c) - cq(c)$. I will first show that incentive compatibility constraints are satisfied if and only if

- i. q is decreasing
- ii. $U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$ for any c

Proof. Using the definition of $U(c)$ we can write the incentive compatibility constraints for any $c' > c$ as follows:

$$\begin{aligned} U(c) &\geq U(c') + q(c')(c' - c) \\ U(c') &\geq U(c) + q(c)(c - c') \end{aligned}$$

Equivalently,

$$q(c')(c' - c) \leq U(c) - U(c') \leq q(c)(c' - c)$$

Therefore, $c' > c$ implies that $q(c) \geq q(c')$. Dividing by $c' - c$ and taking the limit as $c' \rightarrow c$, we get

$$\frac{dU(c)}{dc} = -q(c)$$

Therefore

$$U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$$

Suppose now that q is decreasing and $U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$. Let $c' > c$. we have

$$\begin{aligned} U(c) - U(c') &= \int_{\underline{c}}^{c'} q(\tau) d\tau - \int_{\underline{c}}^c q(\tau) d\tau \\ &= \int_c^{c'} q(\tau) d\tau \\ &\geq q(c')(c' - c) \end{aligned}$$

and hence the IC constraints are satisfied. □

Notice that the participation constraint of type $\bar{c}$, i.e., $U(\bar{c}) \geq 0$ implies that

$$U(\underline{c}) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau = U(\bar{c}) \geq 0.$$

Therefore,

$$U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau \geq \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau - \int_{\underline{c}}^c q(\tau) d\tau = \int_c^{\bar{c}} q(\tau) d\tau \geq 0,$$

Where the first inequality is by the participation constraint of type $\bar{c}$ and the second inequality is by the fact that we are integrating a positive function. Thus, in other words, the participation constraint of type $\bar{c}$ implies all the other participation constraints.

Therefore, the problem is equivalent to

$$\begin{aligned} & \max_{q(c), P(c)} \int_{\underline{c}}^{\bar{c}} (B(q(c)) - P(c)) f(c) dc \\ & \text{subject to } U(\bar{c}) \geq 0 \\ & U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau \\ & q \text{ is decreasing} \end{aligned}$$

Using the definition of $U(c)$ and substituting we can write the objective function as

$$\int_{\underline{c}}^{\bar{c}} \left[B(q(c)) - cq(c) - U(\underline{c}) + \int_{\underline{c}}^c q(\tau) d\tau \right] f(c) dc$$

Substituting for $U(\underline{c})$ using

$$U(\underline{c}) = U(\bar{c}) + \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau,$$

we can write the objective function as

$$\begin{aligned} & \int_{\underline{c}}^{\bar{c}} \left[B(q(c)) - cq(c) - U(\bar{c}) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau + \int_{\underline{c}}^c q(\tau) d\tau \right] f(c) dc \\ & = \int_{\underline{c}}^{\bar{c}} \left[B(q(c)) - cq(c) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau \right] f(c) dc - U(\bar{c}) \end{aligned}$$

At a solution, we must have that $U(\bar{c}) = 0$. We need now to simplify the double integral

$$\int_{\underline{c}}^{\bar{c}} \left(\int_{\underline{c}}^{\bar{c}} q(\tau) d\tau \right) f(c) dc.$$

Using integration by parts this is equal to

$$\int_{\underline{c}}^{\bar{c}} F(c) q(c) dc.$$

The problem is therefore

$$\int_{\underline{c}}^{\bar{c}} \left[B(q(c)) - cq(c) - \frac{F(c)}{f(c)} q(c) \right] f(c) dc$$

subject to the monotonicity constraint. As usual, we will ignore the monotonicity constraint and then show that the solution satisfies it. The first order condition for each c is given by

$$B'(q(c)) = c + \frac{F(c)}{f(c)}$$

Notice that at the lowest cost $\underline{c}$, the solution is identical to the first-best as was expected. Now, for the uniform distribution, $F(c) = \frac{c-\underline{c}}{\bar{c}-\underline{c}}$ and $F(c)/f(c) = c - \underline{c}$ so that

$$B'(q(c)) = c + c - \underline{c} = 2c - \underline{c}$$

Therefore

$$q(c) = (B')^{-1} (2c - \underline{c})$$

Now, if $B'(q) = b$ then $((B')^{-1})'(b) = \frac{1}{B''(q)} = \frac{1}{B''((B')^{-1}(b))} < 0$ for any b since $B'' < 0$. Thus, we have that

$$\frac{dq(c)}{dc} = ((B')^{-1})'(2c - \underline{c}) (2) < 0$$

and therefore, q is strictly decreasing.

Notice also that since $B' > 0$, we have that $q > 0$ for every c . Finally, note that we can compute the transfers by first calculating $U(c)$ and then using $P(c) = U(c) + cq(c)$.